



Full Length Article

Analytical modeling, solution and experimental validation of high-velocity impact properties of composite hexagonal auxetic honeycomb cylindrical shells

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ABSTRACT

Both theoretical and experimental studies are performed to investigate the impact properties of composite hexagonal auxetic honeycomb cylindrical shells subjected to an internal high-velocity projectile impact. Firstly, an analytical model of composite cylindrical shells with two fiber-reinforced polymer (FRP) skins and a hexagonal auxetic honeycomb core (HAHC) is built to anticipate the high-velocity impact properties, with the delamination and fracture energy absorption mechanisms of the skin and the core being considered. A strain-rate fitting function method is proposed to determine the material properties of the FRP skins and the core considering the strain-rate effect. Reddy's higher-order shear deformation theory is utilized to define the displacement of any point of the structure. Also, an improved Gibson theory is applied to derive the equivalent elastic moduli and Poisson's ratios of the HAHC. A detailed experimental validation is conducted on such shell specimens based on a high-velocity impact experimental system to validate the analytical model, in which comprehensive error analysis is discussed. Finally, the influence of critical geometric parameters of the projectile and the studied shell on its impact characteristics is evaluated and some crucial conclusions are provided to enhance impact resistance.

1. Introduction

As one of the basic mechanical components, composite cylindrical shells (CCSs) with different forms of reinforced skins and core structures (Hoang et al., 2021; Tien et al., 2021; Ninh et al., 2022, 2023) have been gradually applied in many engineering fields in aerospace, marine, building, automotive, marine, and defense industries due to their good mechanical properties. Among them, a CCS with a hexagonal auxetic honeycomb core (HAHC) is a new type of composite shell with a negative Poisson ratio core, which has received much attention owing to its good strength and stiffness, impact energy absorption, and weight control properties over conventional structures (Mazloomi et al., 2018; Nguyen et al., 2021). Currently, researchers are trying to apply different composite skins and cores with excellent mechanical properties to

fabricate composite cylindrical casings in aero-engine, missile, and rocket components (Vasiliev et al., 2012; Eyvazian et al., 2021). Many of these structures are often exposed to internal impact loads with low and high speeds during their service lives, which likely results in a significant reduction in structural strength, delamination, structural damage, and non-inclusive accidents (Khalili et al., 2014). As a result, to understand the impact dynamics, damage modes, and failure mechanisms of such shell structures, more theoretical simulations, analyses, and experimental studies should be carried out with different impact velocities (Zhou et al., 2008), penetrating types (Ma et al., 2019), impact angles (Su et al., 2015). This is essential to the application of CCSs with HAHC (or HAHC-CCSs) in engineering sectors.

The dynamic behaviors of the CCSs heavily depend on the impact velocity of the projectile body. For the past decades, many publications

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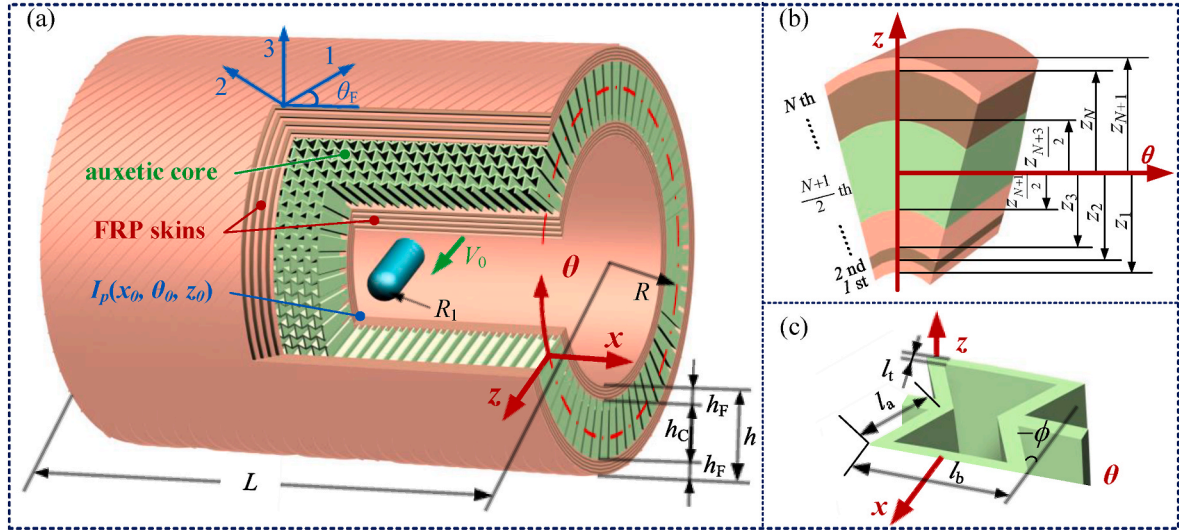


Fig. 1. Configuration of a HAHC-CCS hit by an internal impact projectile: (a) overall structure; (b) enlarged view of $o-z$ plane; (c) enlarged view of a unit cell in the HAHC.

have appeared in the literature on the low-velocity impact studies of different composite shell structures using numerical, analytical, and experimental approaches. For instance, using nine-node degenerated shell elements with assumed shear and membrane strain fields, [Liu and Swaddiwudhipong \(1997\)](#) estimated the low-velocity impact characteristics of composite laminated shells via finite element simulations. Also, the existing calculated and experimental results from the literature were employed to validate the finite element model as well as the prediction results. To simulate the transient response of laminated composite plates and shells impacted by a foreign object, a new shell-based smoothed particle hydrodynamics approach was proposed by [Lin et al. \(2015\)](#). [Liu et al. 2022](#), ([Liu et al., 2017](#)) performed experimental tests on the low-velocity impact parameters and damage modes of a filament-winding composite shell. Also, a finite element (FE) model was developed using a subroutine VUMAT implemented in the commercial code ABAQUS by taking into account the structural intralaminar damage. [Ren et al. \(2018\)](#) experimentally investigated the dynamic buckling of grid-stiffened cylindrical shells made of PEEK 450G subjected to a radial intermediate-velocity impact. They found that with the exponential decay of the incident shock waves, the structural strains were oscillatory and then decreased slowly to positive hoop strains. Under the framework of Reddy's third-order shear deformation theory with the linearized contact law, [Lei et al. \(Lei and Tong, 2019\)](#) successfully obtained the analytical solutions of low-velocity impact properties of composite functionally graded cylindrical shells. They also evaluated the influences of graphene distribution, temperature variation, geometric parameters, and projectile mass on the central deflection of such shells. Based on a proposed finite element model with consideration of Hashin failure criteria, the low-velocity impact parameters of carbon fiber composite curved corrugated sandwich shells were predicted by [Yang et al. \(2020\)](#). Moreover, drop-weight impact tests were conducted to validate the model as well as evaluate the structural impact resistance and failure mechanisms. Based on Love's thin shell theory, the linearized Hertzian contact law, and the Galerkin procedure, the nonlinear impact parameters of corrugated cylindrical shells and functionally graded porous sandwich cylindrical shells were analyzed by [Liu and his co-workers \(Liu et al., 2020, 2022\)](#), in which two novel analytical models were developed, and several practical computational techniques were adopted to derive the nonlinear impact solutions.

Unfortunately, there are only limited studies ([Jing et al., 2013, 2014; Khaire et al., 2022; Tiwari and Khaire, 2022](#)) involving the high-velocity impact properties of metal shells or CCSs with different types of core

structures, in which due to high adaptability and being easy to work with, the finite element approach is primarily employed to forecast and analyze the dynamic behaviors of such shell structures. Here, different high-velocity impact damages and failure modes of constituent materials or elements are taken into account. By using LS-DYNA software, the dynamic responses of metallic cylindrical sandwich shells with an aluminum foam core under high-velocity impact loads were predicted by [Jing and his co-authors \(Jing et al., 2013\)](#). Also, the team ([Jing et al., 2014](#)) fabricated adequate shell specimens to perform the experimental tests based on a gas gun test system. The calculated projectile impact results were validated, and the energy absorption and failure mechanism of such sandwich shells were evaluated via a combination of experimental and theoretical data. By adopting the Johnson-Cook model for the face sheets and the ductile damage material model for the core and adhesive layers, [Khaire and his co-authors \(Khaire et al., 2022; Tiwari and Khaire, 2022\)](#) created a finite element model of isotropic cylindrical sandwich shells with an aluminum honeycomb core to predict the energy absorption and ballistic behavior with the help of Abaqus/PDE module and Abaqus/explicit solver. They also performed a series of experimental tests to validate the model by employing a high-pressurized pneumatic gun with a velocity ranging from 60 to 145 m/s.

To date, limited research work has been devoted to the analytical modeling and prediction of the high-velocity impact characteristics of the CCSs, and the related reports on CCSs with various honeycomb cores are even rarer. To fill this knowledge gap, an analytical model of the HAHC-CCSs under internal high-velocity impact is established in Section 2. Moreover, a strain-rate fitting function method is proposed, with the basic theoretical formulas being derived. Also, the equivalent material properties of the HAHC are determined, and reasonable failure criteria are adopted. Then, the high-velocity impact parameters with consideration of delamination and fracture energy absorption mechanisms are solved in Section 3. A series of specimen preparations, experimental setups, and tests are carried out in Section 4, where the tested and calculated data are compared to give a solid validation of the model. Additionally, the influence of critical geometric parameters of the projectile and the shell structures studied on the impact characteristics is evaluated in Section 5, with some conclusions being summarized in Section 6. The innovation points of this paper are: (1) a novel analytical model of lightweight composite cylindrical shells with an HAHC is developed to investigate the high-velocity impact parameters with the strain rate effect being considered; (2) the impact damage mechanism of

such composite shells subjected to an internal high-velocity projectile impact is revealed; and (3) a preparation flowchart for such composite shell specimens is summarized and practical recommendations are provided for better enhancing their impact resistance. The related modeling and problem-solving approaches used in this study can be easily applied to the high-velocity impact evaluation of other sophisticated composite cylindrical shells with different functional cores, such as honeycomb core, foam core, and truss core.

2. Modelling

2.1. Model description

To predict of the high-velocity impact parameters of a HAHC-CCS structure hit by an internal impact projectile, an analytical model is developed in Fig. 1(a). Suppose that the studied shell includes two fiber-reinforced polymer (FRP) skins and a HAHC structure, in which the inner and outer FRP skins have the same thickness. Based on the findings in Refs. (Mazloomi et al., 2018; Nguyen et al., 2021), the HAHC in such a structure has good impact energy absorption capability with high specific stiffness and strength due to the unique negative Poisson ratio property. Moreover, assuming that a global coordinate system $O - x\theta z$ is located at the middle surface of the core, in which L and h are the overall length and thickness, R is the principal radius of the middle surface, and h_F and h_C are the thickness of each FRP skin and the HAHC. The projectile is supposed to be a hemispherical head cylinder with a radius of R_1 , an initial impact velocity of V_0 , and an impact point of $I_P(x_0, \theta_0, z_0)$. There is also a local coordinate system defined for each FRP skin, with '1', '2', and '3' being the principal material axes and θ_F being the layout angle between the '1' direction and the x -axis. In Fig. 1(b), each FRP layer of the skins or the HAHC is located between the inner surface with a coordinate of z_N and the outer surface with a coordinate z_{N+1} along the thickness direction. The core is assumed to be sandwiched between $z_{(N+1)/2}$ and $z_{(N+3)/2}$, whereas each FRP layer is located within other thickness coordinates. Besides, the schematic of a unit cell in the HAHC is shown in Fig. 1(c), in which l_a is the length of the inclined cell rib, l_b is the length of the horizontal cell rib, l_t is the rib thickness, and ϕ is the inclined angle of such a unit cell.

In addition, to obtain the analytical solutions of the impact parameters of the structure examined, the following basic assumptions are adopted: (1) all FRP layers in the skins and the core are fully bonded without relative slippage (Li et al., 2021); (2) projectile deformation is neglected (Rawat et al., 2017); (3) impact contact area of the shell structure is relatively small and thus can be treated as a circle region with the contact point being located at the center (Lin and Fatt, 2006; Zhang et al., 2013); (4) the related deflection of the structure beyond the impact contact area is small so that this deflection area is assumed to be undamaged; (5) the micro inertial effect and the air compression effect of the core are ignored (Liu et al., 2023); (6) energy consumption related to friction effect in impact process is not considered.

2.2. Material property considering the strain-rate effect

Since the high-velocity impact dynamics of composite honeycomb cylindrical shells are studied in this paper, the strain-rate effect must be taken into account for the proposed model in Section 2.1. By performing dynamic tensile, compression, and shear experiments, Salas (Salas et al., 2009) has discovered that there is an approximately logarithmic relationship between the elastic moduli (including Young's and shear moduli) of the FRP and strain-rate variable. Others (HARDING and WELSH, 1983; Bouette et al., 1992) have also pointed out that the impact of the strain-rate effect on the elastic moduli of the FRP along the in-plane shear and fiber longitudinal direction can be ignored. In addition, as the HAHC in the studied structure is made of an isotropic material, e.g., metal, nylon, or polymer material, the logarithmic function

model from Johnson Cook (Shen et al., 2022) and the power function model adopted by Cowper Symonds (Joćbalis et al., 2023) are usually employed to describe the stress-strain relationships under high-velocity impact loading. However, due to the limitation of respective function equations, it is difficult to accurately determine the stress-strain relationship within a specific strain-rate range (Prasannavenkadesan and Pandithévan, 2021). Therefore, in this work, a strain-rate fitting function method is proposed to define the material properties of the FRP skins and the HAHC, including Young's modulus E_2^S perpendicular to the fiber direction and in-plane shear modulus G_{12}^S of the FRP skins and the related elastic moduli of the isotropic material in the HAHC.

Firstly, for the FRP skins, the elastic moduli under the quasi-static condition are employed to define E_2^S and G_{12}^S when the strain-rate issue is considered, which have the following expressions (Salas et al., 2009):

$$E_2^S = E_2^Q - A_2 E_2^Q \ln(\dot{\epsilon}_2 / \dot{\epsilon}_0) + P_2 \quad (1a)$$

$$G_{12}^S = G_{12}^Q - A_{12} G_{12}^Q \ln(\dot{\epsilon}_{12} / \dot{\epsilon}_0) + P_{12} \quad (1b)$$

where E_2^Q and G_{12}^Q are the quasi-static Young's modulus and in-plane shear modulus; $\dot{\epsilon}_i$ ($i = 2, 12$) represent the strain-rate component in the corresponding direction; $\dot{\epsilon}_0$ represents the reference strain rate; A_i are the fitting coefficients determined by the experimental data; P_i is an n -degree polynomial function that can be fitted based on the measured high-velocity impact data. Here, P_i has the following expression (Shen et al., 2022):

$$P_i = a_{in}(\dot{\epsilon}_i / \dot{\epsilon}_0)^n + a_{in-1}(\dot{\epsilon}_i / \dot{\epsilon}_0)^{n-1} + \dots + a_{i0} \quad (3)$$

where $a_{in}, a_{in-1}, \dots, a_{i0}$ are the polynomial coefficients when a polynomial fitting method is adopted.

Then, by combing the influence of the exponential and logarithmic functions on the strain rate issue of an isotropic material, the Young's modulus E^C and shear modulus G^C of such a raw material that is adopted to create the HAHC in the studied structure when the strain-rate effect is considered can be defined as:

$$E^C = A_E E^Q [1 + B_E \ln(\dot{\epsilon} / \dot{\epsilon}_0) + C_E (\dot{\epsilon} / \dot{\epsilon}_0)^{D_E}] \quad (3a)$$

$$G^C = A_G G^Q [1 + B_G \ln(\dot{\epsilon} / \dot{\epsilon}_0) + C_G (\dot{\epsilon} / \dot{\epsilon}_0)^{D_G}] \quad (3b)$$

where E^Q and G^Q are the quasi-static Young's and shear moduli of the isotropic material that is employed to fabricate the HAHC; $\dot{\epsilon}$ is the strain-rate; $A_E, B_E, C_E, D_E, A_G, B_G, C_G$, and D_G are the fitting coefficients that can be determined based on the measured high-velocity impact data of such a raw material.

2.3. Displacement field and constitutive relation

Based on Reddy's high-order shear deformation theory (Amabili and Reddy, 2020; Yadav et al., 2021), the displacements u, v, w in the x, θ, z directions of any point in the structure investigated can be expressed as:

$$\begin{aligned} u &= u_0 + z\phi_1 - \frac{4}{3h^2}z^3 \left(\phi_1 + \frac{\partial w_0}{\partial x} \right), \\ v &= (1 + z/R)v_0 + z\phi_2 - \frac{4}{3h^2}z^3 \left(\phi_2 + \frac{\partial w_0}{\partial \theta} \right), \\ w &= w_0 \end{aligned} \quad (5)$$

where u_0, v_0, w_0 are the displacements at the middle surface; ϕ_1 and ϕ_2 represent the rotations of the normal to the middle surface with respect to the x and θ directions.

According to the Love-Kirchhoff hypothesis, the strains at any point of the shell structure with consideration of Novozhilov nonlinearity can be defined as (Ben-Youssef et al., 2022):

$$\begin{aligned}
\varepsilon_x &= \frac{1}{2} \left(\frac{\partial w_0}{\partial x} \right)^2 + z \frac{\partial \phi_1}{\partial x} - \frac{4}{3h^2} z^3 \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right), \\
\varepsilon_\theta &= \frac{w_0}{R} + \frac{1}{2} \left(\frac{\partial w_0}{\partial \theta} \right)^2 + z \frac{\partial \phi_2}{\partial \theta} - \frac{4}{3h^2} z^3 \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right), \\
\gamma_{x\theta} &= \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} + z \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) - \frac{8}{3h^2} z^3 \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right), \\
\gamma_{xz} &= \frac{\partial w_0}{\partial x} + \phi_1 - \frac{4}{h^2} z^2 \left(\frac{\partial w_0}{\partial x} + \phi_1 \right), \\
\gamma_{\theta z} &= \frac{\partial w_0}{\partial \theta} + \phi_2 - \frac{4}{h^2} z^2 \left(\frac{\partial w_0}{\partial \theta} + \phi_2 \right)
\end{aligned} \quad (6)$$

The relation expression of stress and strain of the HAHC-CCS in the local coordinate system can be determined as:

$$\sigma^k = \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_{23} \\ \tau_{13} \\ \tau_{12} \end{bmatrix}^k = \begin{bmatrix} Q_{11} & Q_{12} & Q_{13} & 0 & 0 & 0 \\ Q_{21} & Q_{22} & Q_{23} & 0 & 0 & 0 \\ Q_{31} & Q_{32} & Q_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{66} \end{bmatrix}^k \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \gamma_{23} \\ \gamma_{13} \\ \gamma_{12} \end{bmatrix}^k = \mathbf{C}^k \boldsymbol{\varepsilon}^k \quad (k=S, C) \quad (7)$$

where superscript S and C represent the skins and the core, respectively; σ^k and $\boldsymbol{\varepsilon}^k$ are the stress and strain vectors; σ_i ($i=1, 2, 3$) and τ_j ($j=23, 13, 12$) are the normal and shear stress components; ε_i and γ_j are the normal and shear strain components; $\mathbf{C}^k$ is the elastic matrix, in which the stiffness coefficients Q_{mn}^k ($m, n=1, 2, 3, 4, 5, 6$) can be defined as:

$$\begin{aligned}
Q_{11}^k &= \frac{E_1^k (1 - \nu_{23}^k)}{1 - \nu_{23}^k - 2\nu_{12}^k \nu_{21}^k}, Q_{12}^k = Q_{21}^k = Q_{13}^k = Q_{31}^k = \frac{\nu_{12}^k E_2^k}{1 - \nu_{23}^k - 2\nu_{12}^k \nu_{21}^k}, Q_{22}^k = Q_{33}^k = \frac{(1 - \nu_{12}^k \nu_{21}^k) E_2^k}{(1 + \nu_{23}^k)(1 - \nu_{23}^k - 2\nu_{12}^k \nu_{21}^k)}, \\
Q_{23}^k &= Q_{32}^k = \frac{(\nu_{23}^k + \nu_{12}^k \nu_{21}^k) E_2^k}{(1 + \nu_{23}^k)(1 - \nu_{23}^k - 2\nu_{12}^k \nu_{21}^k)}, Q_{44}^k = G_{23}^k, Q_{55}^k = G_{13}^k, Q_{66}^k = G_{12}^k, \nu_{mn}^k E_m^k = \nu_{nm}^k E_n^k
\end{aligned} \quad (8)$$

where E_1^S and E_2^S represent the Young's moduli of two composite skins in '1' and '2' directions, in which E_2^S should be modified from Eq. (1a) to consider the strain-rate effect of FRP; G_{12}^S , G_{13}^S , and G_{23}^S denote the shear moduli in '1-2', '1-3', and '2-3' planes, respectively, in which G_{12}^S also needs to be modified from Eq. (1b) to take into account the related strain-rate effect; ν_{12}^S and ν_{23}^S are the Poisson's ratios induced by the stress in the '1-2' and the '2-3' directions; E_1^C , E_2^C , G_{12}^C , G_{13}^C , G_{23}^C , ν_{12}^C , ν_{21}^C , and ν_{23}^C are the corresponding equivalent Young's and shear moduli and Poisson's ratios of the HAHC, which can be determined by the improved Gibson theory (Gupta and Pradyumna, 2022). Their detailed expressions are as follows:

$$\begin{aligned}
E_1^C &= E^C \frac{b_0^3 (a_0 - \sin \phi)}{\cos^3 \phi [1 + (\tan^2 \phi + a_0 \sec^2 \phi) b_0^3]}, G_{23}^C = G^C \frac{b_0 \cos \phi}{a_0 - \sin \phi}, \nu_{23}^C = \nu^C, \\
E_2^C &= E^C \frac{b_0^3}{\cos \phi (a_0 - \sin \phi) (\tan^2 \phi + b_0^3)}, G_{12}^C = G^C \frac{b_0^3}{a_0 \cos \phi (1 + 2a_0)}, \nu_{12}^C = -\frac{\sin \phi (1 - a_0^2) (a_0 - \sin \phi)}{\cos^2 \phi [1 + (\tan^2 \phi + a_0 \sec^2 \phi) b_0^3]}, \\
E_3^C &= E^C \frac{8}{3\sqrt{3}} b_0, G_{13}^C = G^C \frac{b_0}{2 \cos \phi} \left[\frac{a_0 - \sin \phi}{1 + 2a_0} + \frac{a_0 + 2 \sin^2 \phi}{2(a_0 - \sin \phi)} \right], \nu_{21}^C = -\frac{\sin \phi (1 - a_0^2)}{(\tan^2 \phi + a_0^2) (b_0 - \sin \phi)}
\end{aligned} \quad (9)$$

where ν^C represent the Poisson's ratios of the original material; a_0 and b_0 are the geometric parameters of the HAHC with the following expressions:

$$a_0 = \frac{l_b}{l_a}, b_0 = \frac{l_t}{l_a} \quad (10)$$

According to the volumetric method, the equivalent density of HAHC can be obtained as:

$$\rho^C = \rho_0^C \frac{b_0(a_0 + 2)}{2 \cos \phi (a_0 + \sin \phi)} \quad (10a)$$

where ρ_0^C represents the density of the original material.

2.4. Failure criteria of the FRP skins and the HAHC

For the FRP skins, the three-dimensional Hashin criterion (Ait Mohammed and Tarfaoui, 2022) is used to determine whether the impact damage is started, which is made up of four distinct failure modes and considers the influence of different failure modes and shear effects of the fiber and matrix. For fiber tension ($\sigma_1 \geq 0$) and compression ($\sigma_1 < 0$), one has:

$$\left(\frac{\sigma_1}{X_T^S} \right)^2 + \left(\frac{\tau_{12}}{S_{12}^S} \right)^2 + \left(\frac{\tau_{31}}{S_{13}^S} \right)^2 \geq 1, \sigma_1 \geq 0 \quad (11a)$$

$$\left(\frac{\sigma_1}{X_C^S} \right)^2 \geq 1, \sigma_1 < 0 \quad (11b)$$

For matrix tension ($\sigma_2 + \sigma_3 \geq 0$) and compression ($\sigma_2 + \sigma_3 < 0$), one has:

$$\left(\frac{\sigma_2 + \sigma_3}{Y_T^S} \right)^2 + \frac{\tau_{23}^2 - \sigma_2 \sigma_3}{S_{23}^S{}^2} + \left(\frac{\tau_{12}}{S_{12}^S} \right)^2 + \left(\frac{\tau_{13}}{S_{13}^S} \right)^2 \geq 1, \sigma_2 + \sigma_3 \geq 0 \quad (12a)$$

$$\begin{aligned}
&\frac{\sigma_2 + \sigma_3}{Y_C^S} \left[\left(\frac{Y_C^S}{2S_{23}^S} \right)^2 - 1 \right] + \frac{(\sigma_2 + \sigma_3)^2}{4S_{23}^S{}^2} + \frac{\tau_{23}^2 - \sigma_2 \sigma_3}{S_{23}^S{}^2} + \left(\frac{\tau_{12}}{S_{12}^S} \right)^2 + \left(\frac{\tau_{13}}{S_{13}^S} \right)^2 \geq 1, \sigma_2 \\
&+ \sigma_3 < 0
\end{aligned} \quad (12b)$$

where X_T^S , X_C^S and Y_T^S , Y_C^S denote the tensile and compressive strengths in the '1' and '2' direction of the FRP skins, respectively; S_j^S denote the shear strengths in the three different planes of the FRP skins.

For the HAHc, the Besant failure criterion (Besant et al., 2001) verified by Refs. (Feng and Aymerich, 2013; Liu et al., 2015) is chosen to describe the impact failure, which is one of the most important failure modes for multicellular structure:

$$\left(\frac{\sigma_{33}}{Z_T^C}\right)^2 + \left(\frac{\tau_{13}}{S_{13}^C}\right)^2 + \left(\frac{\tau_{23}}{S_{23}^C}\right)^2 \geq 1 \quad (13)$$

where Z_T^C , S_{13}^C , and S_{23}^C are the corresponding equivalent tensile and shear strength of the HAHc.

2.5. Residual stress distribution and damage evolution

For the l -th layer material that is undergoing a failure process, the progressive failure process from damage initiation to complete failure is described by introducing continuous damage variables d_i^S , d_j^S , and d_{vj}^S .

The total transformed stress vector $\sigma^{(l)}$ at the l -th layer of the HAHc-CCS structure in the global coordinate system is given by:

$$\sigma^{(l)} = \begin{bmatrix} \sigma_x \\ \sigma_\theta \\ \sigma_z \\ \tau_{\theta z} \\ \tau_{xz} \\ \tau_{x\theta} \end{bmatrix}^{(l)} + \frac{h^{(l)}}{h} \begin{bmatrix} \sigma_x^{(l-1)} Q_{11}/K_{11} \\ \sigma_\theta^{(l-1)} Q_{22}/K_{22} \\ \sigma_z^{(l-1)} Q_{33}/K_{33} \\ \tau_{\theta z}^{(l-1)} Q_{44}/K_{44} \\ {}^{(l-1)}Q_{55}/K_{55} \tau_{x\theta}^{(l-1)} Q_{66}/K_{66} \\ \tau_{xz} \end{bmatrix} = \bar{Q}^{(l)} \epsilon^{(l)} + \bar{\sigma}^{(l)} \quad (16)$$

where $\bar{\sigma}^{(l)}$ is the residual stress vector at the l -th layer; $\sigma_j^{(l)}$ ($j = x, \theta, z, \theta z, xz, x\theta$) represents the transformed stress at the l -th layer without considering the distribution of failure stress; $h^{(l)}$ is the thickness of the l -th layer; $\sigma_j^{(l-1)}$ denotes the total transformed stress components of the $l-1$ -th layer; and K_{mn} is the equivalent stiffness of the whole structure, with the following expressions:

$$K_{11} = \frac{\bar{E}_1(1 - \bar{\nu}_{23})}{1 - \bar{\nu}_{23} - 2\bar{\nu}_{12}^2 \frac{\bar{E}_2}{\bar{E}_1}}, K_{22} = K_{33} = K_{11} \frac{\left(1 - \bar{\nu}_{12} \frac{\bar{E}_2}{\bar{E}_1}\right) \bar{E}_2}{(1 + \bar{\nu}_{23})(1 - \bar{\nu}_{23}) \bar{E}_1}, K_{44} = \bar{G}_{23}, K_{55} = \bar{G}_{13}, K_{66} = \bar{G}_{12} \quad (17)$$

According to the Camanho damage model, when the skins meets different damage initiation criteria, the stiffness after damage can be expressed as:

$$dE_i^S = d_i^S E_i^S \quad (14a)$$

$$\begin{aligned} \bar{E}_1 &= \frac{(A_{33}A_{11} - A_{13}^2)(A_{33}A_{22} - A_{23}^2) - (A_{12}A_{33} - A_{13}A_{23})^2}{h(A_{33}A_{22} - A_{23}^2)A_{33}}, \bar{G}_{23} = \frac{A_{44}}{h}, \bar{\nu}_{23} = \frac{A_{23}}{A_{22}} \\ \bar{E}_2 &= \frac{(A_{33}A_{11} - A_{13}^2)(A_{33}A_{22} - A_{23}^2) - (A_{12}A_{33} - A_{13}A_{23})^2}{h(A_{33}A_{11} - A_{23}^2)A_{33}}, \bar{G}_{13} = \frac{A_{55}}{h}, \\ \bar{E}_3 &= \frac{h}{J_0}, \bar{G}_{12} = \frac{(A_{33}A_{11} - A_{13}^2)(A_{11}A_{22} - A_{12}^2) - (A_{11}A_{23} - A_{12}A_{13})^2}{h(A_{11}A_{22} - A_{12}^2)A_{11}}, \bar{\nu}_{12} = \frac{A_{12}}{A_{22}}, \end{aligned} \quad (18)$$

$$dG_j^S = d_j^S G_j^S \quad (14b)$$

$$dv_j^S = d_{vj}^S v_j^S \quad (14c)$$

where dE_i^S , dG_j^S , and dv_j^S are the reduced Young's modulus, shear modulus, and Poisson's ratio after skin damage, respectively. As the damage evolves, when all four damage initiation failure criteria are met simultaneously or the stiffness approaches zero, the layer can be considered as completely failed.

In the global coordinate system, the l -th layer transformed stiffness matrix of the HAHc-CCS structure is stated as:

$$\bar{Q}^{(l)} = T^{(l)T} C^{(l)} T^{(l)} \quad (15)$$

where $T^{(l)}$ stands for the transformation matrix from the material to the global coordinates of the l -th layer, with the detailed expressions being provided in Appendix A.

where $\bar{E}_1$, $\bar{E}_2$, $\bar{G}_{23}$, $\bar{G}_{13}$, $\bar{G}_{12}$, $\bar{\nu}_{12}$ and $\bar{\nu}_{23}$ are the equivalent moduli and Poisson's ratio of the HAHc-CCS structure, respectively, with the detailed expressions being given by:

where $A_{mn} = \sum_{l=1}^N \bar{Q}_{mn}^{(l)} (z_{l+1} - z_l)$ are the tensile stiffness coefficients of the HAHc-CCS structure, and the equations of other corresponding stiffness coefficients B_{mn} , D_{mn} , E_{mn} , F_{mn} , H_{mn} are presented in Appendix B; $J_0 = \sum_{l=1}^N \frac{z_l}{Q_{33}^{(l)}}$ is the coefficients.

3. Solution

3.1. Solution of the displacement and rotation components of the HAHc-CCS structure

The structural displacement functions of the HAHc-CCS structure in the global coordinate system are defined as:

$$\bar{w}^\alpha = \sum_{m'=1}^{M'} \sum_{n'=1}^{N'} \bar{W}_{m'n'}^\alpha P_{m'}^\alpha(\xi) q_{m'}(t) \chi^\alpha \quad (19)$$

where $\bar{w}^\alpha$ ($\alpha = u_0, v_0, w_0, \phi_1, \phi_2$) are the displacement and rotation components; M' and N' are the maximum truncation terms; m' represents the number of polynomial terms, while n' corresponds to the circumferential wave number; $\bar{W}_{m'n'}^\alpha$ are the unknown eigenvectors that need to be solved; $q_{m'}(t)$ are the generalized coordinates; $\chi^{u_0, w_0, \phi_1} = \cos(n'\theta)$ and $\chi^{v_0, \phi_2} = \sin(n'\theta)$; $P_{m'}^\alpha(\xi)$ ($\xi = x/L$) represents the m' -th set of orthogonal polynomials via the Gram-Schmidt process over the interval $[0, 1]$, which can be normalized as:

$$P_{m'}^\alpha(\xi) = \frac{\psi_{m'}^\alpha(\xi)}{\sqrt{\int_0^1 [\psi_{m'}^\alpha(\xi)]^2 d\xi}} \quad (\alpha = u_0, v_0, w_0, \phi_1, \phi_2) \quad (20)$$

$$\iint_A \left\{ \begin{aligned} &\delta u_0 \left[\frac{1}{2} (A_{11} + A_{12}) \left(\frac{\partial w_0}{\partial x} \right)^2 + A_{16} \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} \right] + \delta u_0 \left[B_{11} \frac{\partial \phi_1}{\partial x} + B_{12} \frac{\partial \phi_2}{\partial \theta} + B_{16} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \right] \\ &- \frac{4}{3h^2} \delta u_0 \left[E_{11} \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) + E_{12} \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) + E_{16} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) \right] \end{aligned} \right\} dA = 0 \quad (25a)$$

$$\iint_A \left\{ \begin{aligned} &R \frac{\partial \delta w_0}{\partial x} \left[\left(A_{44} - \frac{4}{h^2} D_{44} \right) \left(\frac{\partial w_0}{\partial x} + \phi_1 \right) + \left(A_{45} - \frac{4}{h^2} D_{45} \right) \left(\frac{\partial w_0}{\partial \theta} + \phi_2 \right) \right] \\ &+ \delta v_0 \left[\frac{1}{2} (A_{12} + A_{22}) \left(\frac{\partial w_0}{\partial x} \right)^2 + A_{26} \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} + B_{12} \frac{\partial \phi_1}{\partial x} + B_{22} \frac{\partial \phi_2}{\partial \theta} + B_{26} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \right] \\ &- \frac{4}{3h^2} \delta v_0 \left[E_{12} \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) + E_{22} \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) + E_{26} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) \right] \end{aligned} \right\} dA = R \delta w_0 F \quad (25b)$$

$$\iint_A \left\{ \begin{aligned} &\frac{\partial \delta \phi_1}{\partial x} \left[\frac{1}{2} (B_{11} + B_{12}) \left(\frac{\partial w_0}{\partial x} \right)^2 + B_{16} \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} + D_{11} \frac{\partial \phi_1}{\partial x} + D_{12} \frac{\partial \phi_2}{\partial \theta} + D_{16} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \right] \\ &- \frac{4}{3h^2} \frac{\partial \delta \phi_1}{\partial x} \left[F_{11} \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) + F_{12} \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) - F_{16} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) \right] \\ &- \delta w_0 \left[\left(A_{44} - \frac{4}{h^2} D_{44} \right) \left(\frac{\partial w_0}{\partial x} + \phi_1 \right) + \left(A_{45} - \frac{4}{h^2} D_{45} \right) \left(\frac{\partial w_0}{\partial \theta} + \phi_2 \right) \right] \end{aligned} \right\} dA = 0 \quad (25c)$$

where the leading terms of the polynomials $\psi_{m'}^\alpha(\xi)$ have different expressions, such as $\psi_1^u(\xi) = \psi_1^v(\xi) = \xi^2 - \xi$ and $\psi_1^w(\xi) = \psi_1^{\phi_1}(\xi) = \psi_1^{\phi_2}(\xi) = \xi^4 - 2\xi^3 + \xi^2$ to satisfy the clamped boundary conditions. Also, the following terms can be recursively obtained through the equations below:

$$\begin{aligned} \psi_2^\alpha(\xi) &= (\xi - \eta_{m'}^\alpha) \psi_1^\alpha(\xi) \\ \psi_{m'}^\alpha(\xi) &= (\xi - \eta_{m'}^\alpha) \psi_{m'-1}^\alpha(\xi) - \mu_{m'}^\alpha \psi_{m'-2}^\alpha(\xi), m' \geq 2 \end{aligned} \quad (21)$$

where $\eta_{m'}$ and $\mu_{m'}$ have the following expression when the weight function is set to 1:

$$\eta_{m'}^\alpha = \frac{\int_0^\xi [\psi_{m'-1}^\alpha(\xi)]^2 \xi d\xi}{\int_0^\xi [\psi_{m'-1}^\alpha(\xi)]^2 d\xi} \quad (22a)$$

$$\mu_{m'}^\alpha = \frac{\int_0^\xi [\psi_{m'-1}^\alpha(\xi)]^2 [\psi_{m'-2}^\alpha(\xi)]^2 \xi d\xi}{\int_0^\xi [\psi_{m'-2}^\alpha(\xi)]^2 d\xi}, m' \geq 2 \quad (22b)$$

The displacement equations at the impact center of the HAHCCS structure can be defined as:

$$w_0(x_0, \theta_0) = w_0(x_0 + R_2, \theta_0) = w_0(x_0, \theta_0 + \theta_2) = w_{\max} \quad (23a)$$

$$\phi_1(x_0, \theta_0) = \phi_1(x_0 + R_2, \theta_0) = \phi_1(x_0, \theta_0 + \theta_2) = 0 \quad (23b)$$

$$\phi_2(x_0, \theta_0) = \phi_2(x_0 + R_2, \theta_0) = \phi_2(x_0, \theta_0 + \theta_2) = 0 \quad (23c)$$

where $w_{\max}$ is the maximum displacement of the structure at the impact point; R_2 stands for the relatively small radius of the contact area; θ_2 is the arc length of the contact area that can be easily obtained by the Pythagorean theorem:

$$\theta_2 = \arccos \frac{R^2 + (R - R_1 + w_{\max})^2 - R_1^2}{2R(R - R_1 + w_{\max})}, R_2 = R \sin \theta_2 \quad (24)$$

According to the principle of virtual work, by substituting the expressions of generalized internal force and moment (Nguyen et al., 2020), the governing differential equations (Jones, 1997) are deduced as:

where δu_0 , δv_0 , δw_0 , $\delta \phi_1$, and $\delta \phi_2$ denote the virtual variations of u_0 , v_0 , w_0 , ϕ_1 , and ϕ_2 ; the differential element of the middle surface of the HAHCCS structure is defined as $dA = R(1+z/R)dx d\theta$ (Leissa and Chang, 1996); F is the contact force. Using the modified Hertzian contact law proposed by Choi (Choi and Hong, 1994), F can be expressed as:

$$F = \frac{4\sqrt{R_1}(w_{\max})^{3/2}}{3(1 - \nu_{\text{im}}^2)/E_{\text{im}} + 1/E_2^{\text{up}}} \quad (26)$$

where ν_{im} and E_{im} are the Poisson's ratio and Young's modulus of the impactor; E_2^{up} is the Young's modulus of the innermost layer of the undamaged structure in the '2' direction.

Substituting Eqs. (23) to (25) into Eq. (19), the unknown eigenvectors $\bar{W}_{m'n'}^\alpha$ and the corresponding displacement components $\bar{w}^\alpha$ can be calculated.

3.2. Solution of the impact velocity of the projectile with the innermost layer being perforated

The total strain energy U of the HAHCCS structure as referenced from Ref. (Khalili et al., 2012) can be determined by considering various strain energies, including the energies associated with tension,

tension-bending, bending, tension-torsion, bending-torsion, and torsion effects, namely U_A , U_B , U_D , U_E , U_F , and U_H . They can be acquired in terms of the displacement and rotation w_0 , ϕ_1 , and ϕ_2 by introducing the strain-displacement relationships given in Eq. (5) with the following

$$U_A = \iint_A \left\{ \begin{aligned} & A_{11} \frac{1}{4} \left(\frac{\partial w_0}{\partial x} \right)^4 + 2A_{12} \left[\frac{1}{2} \frac{w_0}{R} \left(\frac{\partial w_0}{\partial x} \right)^2 + \frac{1}{4} \left(\frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} \right)^2 \right] + 2A_{16} \left[\frac{1}{2} \left(\frac{\partial w_0}{\partial x} \right)^2 \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} \right] \\ & + A_{22} \left[\left(\frac{w_0}{R} \right)^2 + \frac{1}{4} \left(\frac{\partial w_0}{\partial \theta} \right)^4 + \frac{w_0}{R} \left(\frac{\partial w_0}{\partial \theta} \right)^2 \right] + 2A_{26} \left[\frac{w_0}{R} \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} + \frac{1}{2} \left(\frac{\partial w_0}{\partial \theta} \right)^2 \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} \right] \\ & + A_{66} \left(\frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} \right)^2 + A_{44} \left[\left(\frac{\partial w_0}{\partial \theta} \right)^2 + \phi_2^2 + 2 \frac{\partial w_0}{\partial \theta} \phi_2 \right] + 2A_{45} \left[\frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} + \phi_2 \frac{\partial w_0}{\partial x} + \phi_1 \frac{\partial w_0}{\partial \theta} + \phi_1 \phi_2 \right] \\ & + A_{55} \left[\left(\frac{\partial w_0}{\partial x} \right)^2 + \phi_1^2 + 2 \frac{\partial w_0}{\partial x} \phi_1 \right] \end{aligned} \right\} dA \quad (27a)$$

$$U_B = \iint_A \left\{ \begin{aligned} & B_{11} \frac{\partial \phi_1}{\partial x} \left(\frac{\partial w_0}{\partial x} \right)^2 + 2B_{12} \left\{ \left[\frac{w_0}{R} + \frac{1}{2} \left(\frac{\partial w_0}{\partial \theta} \right)^2 \right] \frac{\partial \phi_1}{\partial x} + \frac{1}{2} \left(\frac{\partial w_0}{\partial x} \right)^2 \frac{\partial \phi_2}{\partial \theta} \right\} \\ & + 2B_{16} \left[\frac{\partial \phi_1}{\partial x} \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} + \frac{1}{2} \left(\frac{\partial w_0}{\partial x} \right)^2 \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \right] + B_{22} \left[2 \frac{w_0}{R} \frac{\partial \phi_2}{\partial \theta} + \frac{\partial \phi_2}{\partial \theta} \left(\frac{\partial w_0}{\partial \theta} \right)^2 \right] \\ & + 2B_{26} \left[\frac{w_0}{R} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) + \frac{\partial \phi_2}{\partial \theta} \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} + \frac{1}{2} \left(\frac{\partial w_0}{\partial \theta} \right)^2 \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \right] \\ & + B_{66} \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \end{aligned} \right\} dA \quad (27b)$$

$$U_D = \iint_A \left\{ \begin{aligned} & D_{11} \left(\frac{\partial \phi_1}{\partial x} \right)^2 + 2D_{12} \left(\frac{\partial \phi_1}{\partial x} \frac{\partial \phi_2}{\partial \theta} \right) + 2D_{16} \left[\frac{\partial \phi_1}{\partial x} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \right] + 2D_{26} \left[\frac{\partial \phi_2}{\partial \theta} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \right] \\ & + D_{22} \left(\frac{\partial \phi_2}{\partial \theta} \right)^2 + D_{66} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right)^2 + D_{44} \left[\frac{-8\phi_2}{h^2} \left(\frac{\partial w_0}{\partial \theta} + \phi_2 \right) + \frac{-8}{h^2} \frac{\partial w_0}{\partial \theta} \left(\frac{\partial w_0}{\partial \theta} + \phi_2 \right) \right] \\ & + 2D_{45} \left[\frac{-4}{h^2} \frac{\partial w_0}{\partial x} \left(\frac{\partial w_0}{\partial \theta} + \phi_2 \right) + \frac{-4\phi_1}{h^2} \left(\frac{\partial w_0}{\partial \theta} + \phi_2 \right) + \frac{-4}{h^2} \frac{\partial w_0}{\partial \theta} \left(\frac{\partial w_0}{\partial x} + \phi_1 \right) + \frac{-4\phi_2}{h^2} \left(\frac{\partial w_0}{\partial x} + \phi_1 \right) \right] \\ & + D_{55} \left[\frac{-8\phi_1}{h^2} \left(\frac{\partial w_0}{\partial x} + \phi_1 \right) + \frac{-8}{h^2} \frac{\partial w_0}{\partial x} \left(\frac{\partial w_0}{\partial x} + \phi_1 \right) \right] \end{aligned} \right\} dA \quad (27c)$$

$$U_E = \frac{-4}{3h^2} \iint_A \left\{ \begin{aligned} & E_{11} \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) \left(\frac{\partial w_0}{\partial x} \right)^2 + E_{22} \left[\left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) \left(\frac{\partial w_0}{\partial \theta} \right)^2 + 2 \frac{w_0}{R} \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) \right] \\ & + 2E_{12} \left[\frac{w_0}{R} \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) + \frac{1}{2} \left(\frac{\partial w_0}{\partial \theta} \right)^2 \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) + \frac{1}{2} \left(\frac{\partial w_0}{\partial x} \right)^2 \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) \right] \\ & + 2E_{16} \left[\left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} + \frac{1}{2} \left(\frac{\partial w_0}{\partial x} \right)^2 \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) \right] \\ & + 2E_{26} \left[\frac{w_0}{R} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) + \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} \right. \\ & \quad \left. + \frac{1}{2} \left(\frac{\partial w_0}{\partial \theta} \right)^2 \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) \right] \\ & + E_{66} \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial \theta} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) \end{aligned} \right\} dA \quad (27d)$$

$$U_F = \iint_A \left\{ \begin{aligned} & F_{11} \frac{\partial \phi_1}{\partial x} \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) + 2F_{12} \left[\frac{\partial \phi_1}{\partial x} \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) + \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) \frac{\partial \phi_2}{\partial \theta} \right] \\ & + 2F_{16} \left[\frac{\partial \phi_1}{\partial x} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) + \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \right] \\ & + 2F_{26} \left[\frac{\partial \phi_2}{\partial \theta} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) + \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) \right] \\ & + 2F_{22} \frac{\partial \phi_2}{\partial \theta} \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) + F_{66} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} \right) \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) \\ & + \frac{16}{h^4} \left\{ F_{44} \left(\frac{\partial w_0}{\partial \theta} + \phi_2 \right)^2 + 2F_{45} \left(\frac{\partial w_0}{\partial x} + \phi_1 \right) \left(\frac{\partial w_0}{\partial \theta} + \phi_2 \right) + F_{55} \left(\frac{\partial w_0}{\partial x} + \phi_1 \right)^2 \right\} \end{aligned} \right\} dA \quad (27e)$$

$$U_H = \frac{16}{9h^4} \iint_A \left\{ \begin{aligned} & H_{11} \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right)^2 + 2H_{12} \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) \\ & + 2H_{16} \left(\frac{\partial \phi_1}{\partial x} + \frac{\partial^2 w_0}{\partial x^2} \right) \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) + H_{22} \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right)^2 \\ & + 2H_{26} \left(\frac{\partial \phi_2}{\partial \theta} + \frac{\partial^2 w_0}{\partial \theta^2} \right) \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right) + H_{66} \left(\frac{\partial \phi_1}{\partial \theta} + \frac{\partial \phi_2}{\partial x} + \frac{2\partial^2 w_0}{\partial x \partial \theta} \right)^2 \end{aligned} \right\} dA \quad (27f)$$

expressions:

When the innermost layer of the HAHCCS structure is perforated by the high-velocity impact projectile, delamination, and fracture damage will occur. According to the force-driven model proposed by Davies, Meng, and their coauthors (Davies et al., 2006; Meng et al., 2018), the delamination considering the cohesive layer failure will propagate in mode II in the mid-plane (Wu et al., 2012). Therefore, the energy consumed by delamination damage can be obtained as:

$$T_d = \frac{\pi(\bar{E}_1 + \bar{E}_2)hG_{II}^2}{4(1 - \bar{V}_{12}^2)\sigma_{II}^2} \quad (28)$$

where G_{II} is the mode II critical inter-laminar energy release rate of the adhesive material that is affected by the maximum displacement at the impact point, the material type, and structural geometries (Huang et al., 2020), with the expression being provided in Appendix C; σ_{II} is the critical inter-laminar shear strength with its value being referenced from the literature (Januš et al., 2022).

Furthermore, the energies consumed by fracture damage of the skins T_f^S (Li et al., 2022) and the core T_f^C (Allahverdizadeh et al., 2008) are determined as:

$$T_f^S = \frac{w_{\max} \pi e_{up}^S R_1^2}{3} \quad (29a)$$

$$T_f^C = \frac{h_c^2 \pi e_{up}^C R_1}{4} \quad (29b)$$

where e_{up}^S and e_{up}^C are the energy density of the innermost layer of the undamaged structure. It should be noted that no matter how the consumed energies in Eqs. (28) and (29) are changed, the primary energy consumptions throughout the process of impact of the HAHCCS structure are still the strain energy consumptions as described in Eq. (27).

Based on the energy balance principle, the impact velocity after the

innermost layer is perforated can be stated as:

$$V = \sqrt{V_0^2 - \frac{2(T_u + T_d + T_f^k)}{M}} \quad (30)$$

where M is the mass of the projectile; T_u represents the consumed strain energy with the expression being shown in Appendix D.

As a result, the impact time t after the innermost layer is perforated can be determined as:

$$t = \frac{[M + m_e]|V - V_0| + C_s w_{\max}}{F - K_{66} w_{\max}} \quad (31)$$

where C_s is the overall viscous damping coefficient given in (Gromysz, 2012; Zelin et al., 2020); m_e denotes the effective mass of the HAHCCS structure, which is obtained by neglecting its rotatory inertia (Morinière et al., 2013) to make the velocity profile similar to the deformation profile of the structure, expressed as:

$$m_e = \pi R_2^2 \rho^{up} + \frac{2}{9}(L - 2R_2) \left(\pi - \frac{R_2}{R} \right) \rho^{up} \quad (32)$$

where ρ^{up} is the density of the innermost layer, which can be taken as the density of the skin ρ^S or the HAHCC ρ^C .

3.3. Iterative solution of impact parameters

After the projectile perforates the N -th layer of the structure, the impact energy absorption EA and the specific energy absorption SEA can be acquired with the following equations (Al Antali et al., 2017):

$$EA = \sum_{l=1}^N \left(T_u^{(l)} + T_d^{(l)} + T_f^{(l)} \right) \quad (33)$$

$$SEA = \frac{EA}{m_s} \quad (34)$$

where $T_u^{(l)}$, $T_d^{(l)}$, and $T_f^{(l)}$ denote the energy consumed by strain energy, delamination damage, and fracture damage with the l -th layer of the structure being perforated; m_s is the mass of the structure.

Additionally, according to the ballistic testing V50 standard described in Ref. (Khair et al., 2021), the ballistic limit V_{bl} of the structure can be determined as:

$$V_{bl} = \frac{V_{\max} + V_{\min}}{2} \quad (35)$$

where $V_{\max}$ is the maximum initial velocity which ensures the non-perforation phenomenon of the structure; $V_{\min}$ denote the minimum initial velocity which results in complete perforation. It should be noted that to solve Eq. (34), the residual velocity $V^{(N)}$ of the projectile after perforating the N -th layer of the structure must be determined using an iterative algorithm, as shown in Fig. 2, which can be realized via a calculation program written based on MATLAB.

The above iterative algorithm described in Fig. 2 can be divided into the following four key steps. Step 1: input the structural and material parameters of the HAHC-CCS structure as well as the initial parameters of the iterative algorithm into the calculation program, which includes the layer counter l , initial displacement $w_{\max}$, initial step size e , step size precision coefficient c_1 , initial strain-rate vector $\dot{\epsilon}^{(0)}$, strain-rate precision coefficient c_2 , and the initial impact velocity. Step 2: calculate the total transformed stress vector $\sigma^{(l)}$ by substituting the displacement $w_{\max}$ into Eqs. (5) to (6). Subsequently, use Eqs. (11) to (13) to determine whether the failure criteria of the FRP skins and the HAHC are satisfied. If these criteria are not met, amplify $w_{\max}$ and continue to perform calculation work. Otherwise, the impact damage phenomenon is regarded to occur in the l -th layer of the HAHC-CCS structure. At this time, use two inequalities ($e < c_1$ and $\dot{\epsilon}^{(l)}/\dot{\epsilon}^{(l-1)} < c_2$) to determine whether e is smaller than c_1 and whether the iterative calculation values related to strain-rate and step size meet the pre-defined accuracy requirement. As long as one of two inequalities fails to meet the related requirement, reduce e by half and modify the values of $E_2^s(\dot{\epsilon})$, $G_{12}^s(\dot{\epsilon})$, $E^c(\dot{\epsilon})$ and $G^c(\dot{\epsilon})$ in Eqs. (1) to (3). Note that in each iterative calculation, the related impact parameters, $T_u^{(l)}$, $T_d^{(l)}$, $T_f^{(l)}$, $V^{(l)}$, and t in Eqs. (27)–(31) will be recorded. Step 3: when the concerned conditions in the above two inequalities are satisfied, let $V_0 = V^{(l)}$ and $l = l + 1$, and delete the l -th layer to obtain a new undamaged structure with $N-1$ layers. Moreover, update the structural and material parameters and conduct the iterative calculation again to determine the impact velocity $V^{(l)}$ of the l -th layer until the layer counter reaches the total number of layers N . Otherwise, it is true in the expression $V^{(l)} \leq 0$, which proves that the HAHC-CCS structure is not perforated. Step 4: solve the concerned value of $V^{(N)}$ of the projectile after the structure is perforated. As a result, V_{bl} can be calculated via Eq. (35). Note that in this step, different initial impact velocities need to be inputted into the calculation program to obtain $V_{\max}$ and $V_{\min}$, since V_{bl} is closely related to $V_{\max}$ and $V_{\min}$.

Once V_{bl} is obtained, the damage areas S_d^{in} and S_d^{out} on the inner and outer skins of such a perforated shell structure can be solved by modifying the calculation formulas of the damaged area described in Ref. (Sundararajan, 1981), which have the following expressions:

$$S_d^{\text{in}} = p^{\text{in}} S_p \frac{V_{bl}^{g^{\text{in}}+s^{\text{in}}}}{V_0^{g^{\text{in}}}} \quad (36a)$$

$$S_d^{\text{out}} = p^{\text{out}} S_p \frac{V_{bl}^{g^{\text{out}}+s^{\text{out}}}}{V_0^{g^{\text{out}}}} \quad (36b)$$

where $S_p = \pi R_2^2$ is the cross-sectional area of the contact area; p^{in} , p^{out} , g^{in} , g^{out} , s^{in} , and s^{out} are the engineering coefficients (Sundararajan, 1981).

4. Experimental validation

4.1. Specimens fabrication

Several HAHC-CCS specimens made of T300/E51 FRP skins and XF 1200F nylon HAHC are fabricated to conduct the high-velocity impact experimental tests with the geometric parameters of the specimens being: $L = 130$ mm, $h = 16$ mm, $R = 70$ mm, $l_a = 5.5$ mm, $l_b = 10.5$ mm, $l_t = 1$ mm, $\phi = -30^\circ$, and the lay-up order being $[0/+45/90/-45/0/C_{12.5}]_S$, in which $C_{12.5}$ represents the HAHC layer with $h_c = 12.5$ mm. With a combined mass of 22 g, the projectile has a hemispherical head and a cylindrical body with dimensions of 20 mm in diameter and 34 mm in length. Table 1 lists the material parameters of the FRP skins and the HAHC. The majority of these parameters are provided by the manufacturers (Yixing Siweiqi Carbon Fiber Products Co., Ltd. China and Shenzhen Pengda 3D Intelligent Manufacturing Technology Co., Ltd., China). A series of stretch tests are conducted by these companies on the related specimens based on the universal testing machine. As a result, the material parameters of the FRP skins and the raw material that is adopted to make the HAHC can be obtained according to the test standard and procedures proposed in the ASTM

Table 1

Material properties of FRP skins and HAHC of the specimens.

Type	FRP skins	Raw material of the core
Young's moduli (GPa)	$E_1^s = 105, E_2^s = E_3^s = 6.5$	$E^c = 1.951$
Shear moduli (GPa)	$G_{12}^s = G_{13}^s = 5.5, G_{23}^s = 2.0$	$G^c = 1.054$
Poisson's ratio	$\nu_{12}^s = \nu_{13}^s = 0.27, \nu_{23}^s = 0.625$	$\nu^c = 0.34$
Tensile strength (GPa)	$X_T^s = 2.39, Y_T^s = 0.025$	$Z_T^c = 0.105$
Compressive strength (GPa)	$X_C^s = 1.67, Y_C^s = 0.15$	–
shear strength (GPa)	$S_{12}^s = S_{23}^s = S_{31}^s = 0.03$	$S_{13}^c = 0.11, S_{23}^c = 0.07$
Density (kg/m ³)	$\rho^s = 1600$	$\rho_o^c = 990$

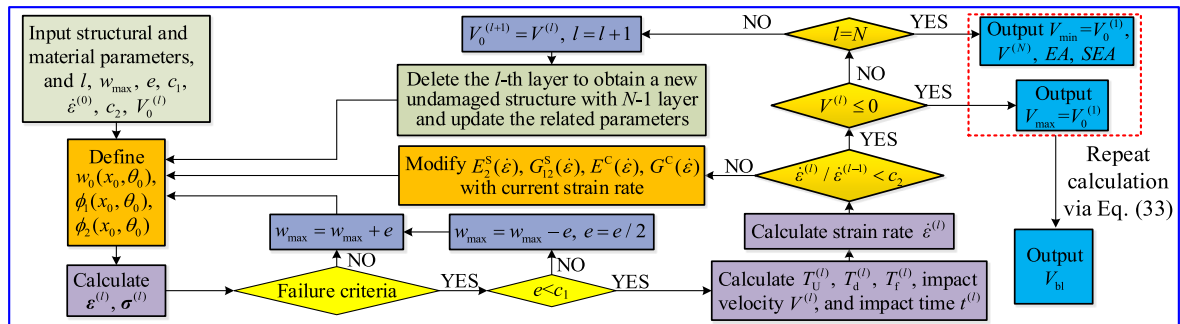


Fig. 2. An iterative solution algorithm of the impact parameters of the HAHC-CCS hit by an internal impact projectile.

Table 2

Fitting coefficients calculated based on the experimental data when the reference strain-rate is set to $\dot{\epsilon}_0 = 1 \text{ s}^{-1}$.

FRP skins	HAHC
$A_2 = -8.37 \times 10^{-3}$, $P_2 = 1.822 \times 10^5 (\dot{\epsilon}_2 / \dot{\epsilon}_0) + 0.8143$	$A_E = 1.005$, $B_E = 0.012$, $C_E = 0.001$, $D_E = 0.912$
$A_{12} = -9.833 \times 10^{-2}$, $P_{12} = 7.009 \times 10^{-2} (\dot{\epsilon}_{12} / \dot{\epsilon}_0)^3 + 0.4505$	$A_G = 1.006$, $B_G = 0.006$, $C_G = 0.0001$, $D_G = 0.961$

D3039M-00. Note that the values of Z_T^C , S_{13}^C , and S_{23}^C are determined by combing the analysis data in Ref. (Calignano et al., 2020).

Here, using the strain rate fitting function method proposed in this study, the fitting coefficient values of FRP skins and the HAHC can be obtained, as listed in Table 2, in which the fitting calculations are based on experimental data in Refs. (Gilat et al., 2002; Richeton et al., 2003; Ha et al., 2023). Also, the initial parameters used in the iterative calculations are: $l = 1$, $w_{\max} = 0$, $e = 0.5$, $c_1 = 0.01$, $\dot{\epsilon}^{(0)} = 10^{-2}$, and $c_2 = 10$. Note that they are also employed in subsequent theoretical calculations. The damage variables under different damage modes are given in Table 3.

A fabrication flowchart of a HAHC-CCS specimen is shown in Fig. 3, in which the following key steps are adopted: (1) Cut the prepreg into desired pieces and place them on the surface of the molds at a pre-determined angle after applying the mold release wax and the peel ply to the molds; (2) Compress the related prepreg pieces via Molds I, II, and III, and tighten the bolts of these molds with a torque of 5 Nm before placing it into the multifunctional heating box for preheating at 80 °C for

Table 3

Damage variables under different damage initiation criteria.

Damage mode	Damage variables
fiber tension	$d_1^S = d_2^S = d_{12}^S = d_{v12}^S = 0.07$
fiber compression	$d_1^S = d_2^S = d_{12}^S = d_{v12}^S = 0.14$
matrix tension	$d_2^S = 0.2$, $d_{12}^S = d_{23}^S = 0.22$, $d_{v12}^S = 0.15$
matrix compression	$d_2^S = d_{12}^S = d_{23}^S = 0.4$, $d_{v12}^S = 0.15$
fiber and matrix tension	$d_1^S = 0.07$, $d_2^S = 0.2$, $d_{12}^S = 0.22$, $d_{v12}^S = 0.15$
fiber and matrix compression	$d_1^S = 0.14$, $d_2^S = 0.4$, $d_{12}^S = 0.4$, $d_{v12}^S = 0.15$

30 min and heating at 150 °C for 30 min to obtain the inner and outer skins; (3) Use a 3D printer to fabricate the HAHC with the high-performance nylon powder; (4) Adopt the adhesive film of epoxy resins (Ec0711, Zhongjin Hongwang (Beijing) Composite Metal Material Co., Ltd., China) to bond the HAHC to the inner and outer skins and put the bonded structure into the Molds I and III before heating it in the multifunctional heating box at 120 °C for 90 min to obtain the HAHC-CCS structure.

4.2. Experimental setup

Fig. 4 illustrates a high-velocity impact experimental system for the HAHC-CCS specimens, which comprises a projectile launching system, a target box, and a velocity measurement system. The projectile launching system consists of several projectiles, a control box, a high-pressure gas cylinder, a delivery pipeline, a launch gas storage device, a pressure-reducing valve, a projectile launch tube, and an inlet valve, as shown in Fig. 4(a). These valves are used to control the compressed gas to enter the gas chamber through the delivery pipeline and a rotary switch is employed to limit the maximum pressure of the gas. The target box is designed to absorb the remaining impact velocity of the projectile with a steel wire with a set of fixtures being adopted to securely clamp the specimen, as shown in Fig. 4(b). Also, this box is embedded with a light-transmitting glass to capture the flight position of the projectile. The velocity measurement system includes a PC, an illuminator, a high-speed camera, and a velocimeter framework. The velocimeter framework has two columns of infrared receiving phototransistor array and infrared emitting phototransistor array for testing the fly velocity of the projectile. The illuminator provides ample luminance inward the target box through a side transparent glass, and the high-speed camera captures the images of different flight paths of the projectile after perforation of each specimen. Therefore, the residual velocity of the projectile can be acquired indirectly. Table 4 lists the detailed instruments used in the test system.

Before starting the experimental test, several steps are taken to ensure the system is properly set up and calibrated (Zhang et al., 2022). First, load the gas from the high-pressure gas cylinder to the gas chamber to control the initial velocity of the projectile by inputting a certain pressure value into the software interface of the control box, as shown in

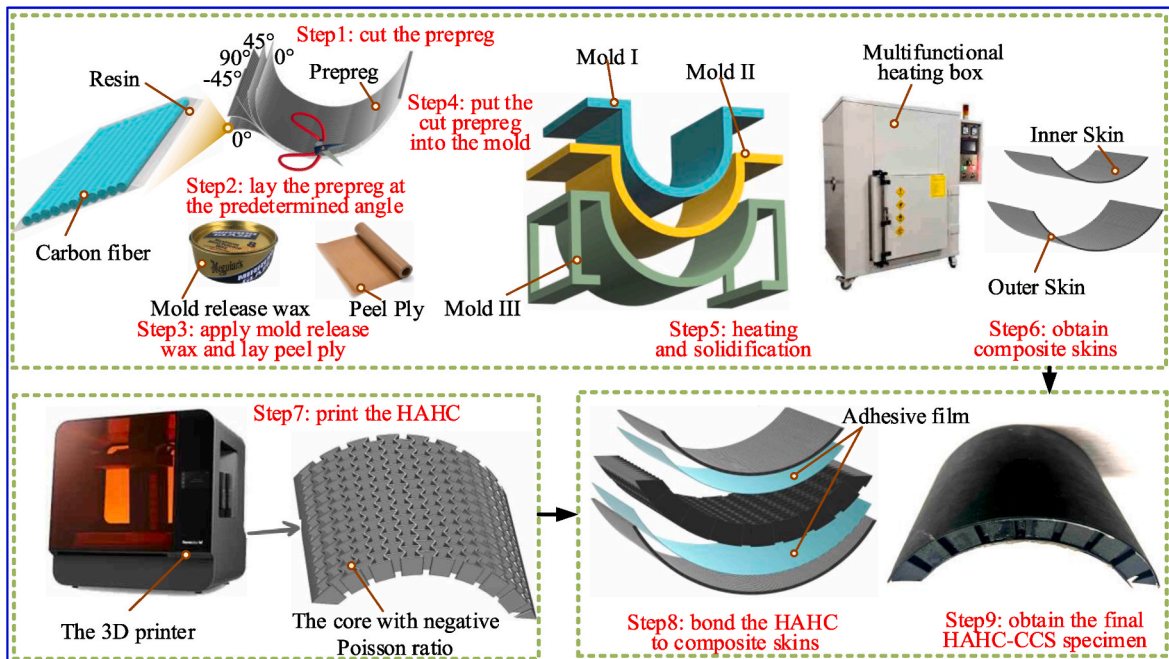


Fig. 3. A flowchart for fabrication of a HAHC-CCS specimen.

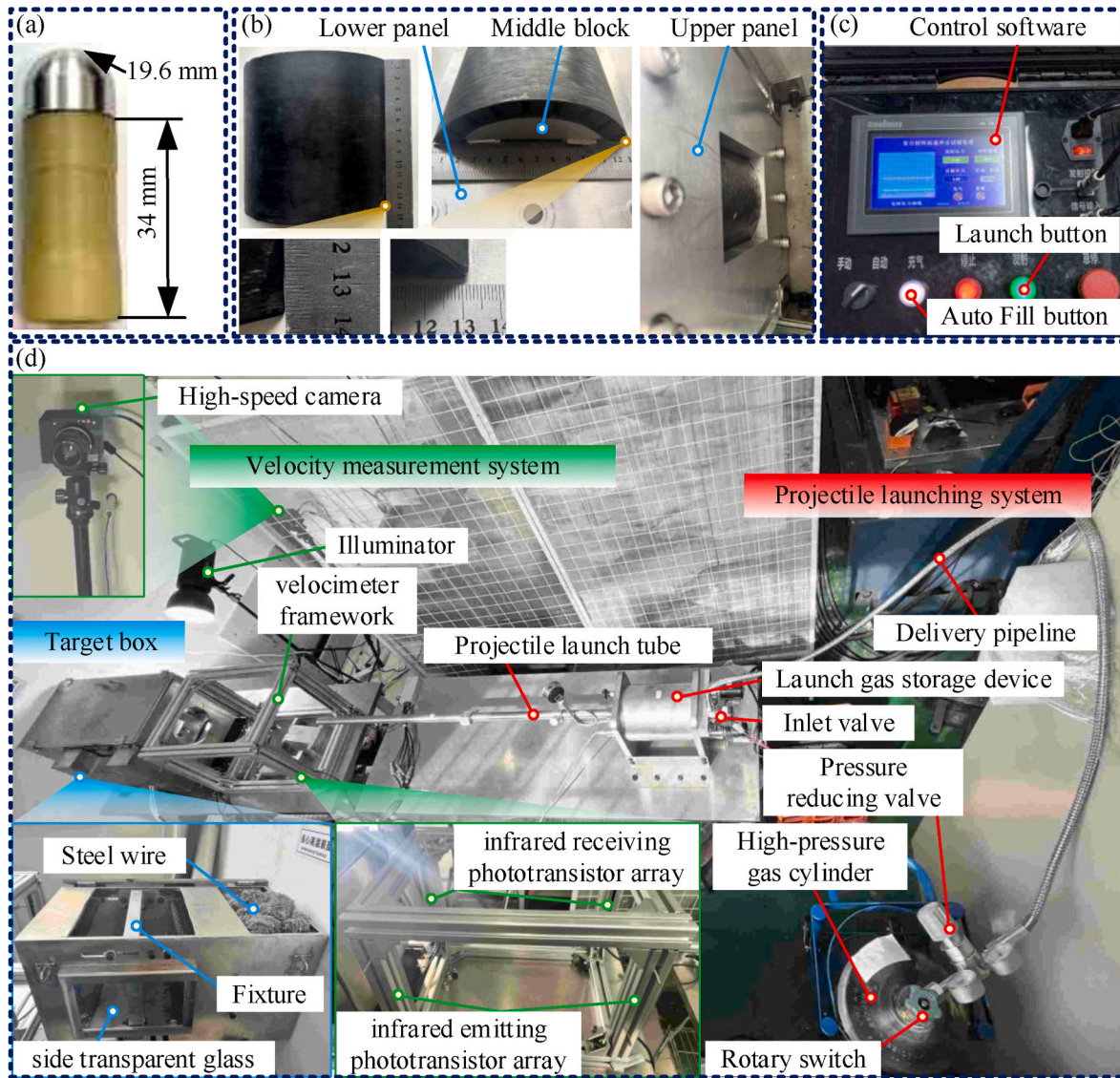


Fig. 4. A high-velocity impact experimental system for the HAHC-CCS specimens: (a) projectile; (b) specimen and fixture; (c) control box; (d) the whole impact experimental system.

Fig. 4(c). Next, clamp the specimen in the fixture at the target box, and place the projectile at the end of the launch tube with its outlet direction being adjusted towards the geometric center of the fixture. Then, power on the velocity measurement system, turn on the illuminator, connect the high-speed camera to the PC, and set the corresponding acquisition parameters. During the experiments, the projectile is launched from the projectile launching system and is rapidly accelerated by the high-intensity gas released from the gas chamber, resulting in the perforation of the specimen. As the projectile passes through the phototransistors, the initial velocity is recorded. The residual velocity can be obliquely measured by calculating the fighting distance captured in the continuous high-speed photographs. These experiments are repeated at various initial velocities to determine the relationship between the initial and residual velocities of the projectiles.

4.3. Comparison of the experimental and predicted results

Based on the fabrication method proposed in Section 4.3.1, five identical specimens with the name of Specimen I to V are manufactured and tested to provide a validation of the established model in solving various high-velocity impact parameters. Using the high-velocity impact

experimental system established in Section 4.3.2, the tests are conducted on these specimens at various initial impact velocities of 101.9, 116.9, 140.1, 147.3, and 162.0 m/s respectively, which can be collected by the velocimeter framework. By taking Specimens III, IV, and V as examples, Fig. 5 displays the fighting distance of the projectile between the continuous photographs measured by the high-speed camera as the penetration damage occurs on each specimen with various initial velocities. Here, the frame rate adopted is 1984 FPS. In this way, all the residual velocities of the projectile at different initial impact velocities can be obtained indirectly by multiplying the flight distance with the frame rate, as shown in Fig. 6. Also, Fig. 6 compares the experimental and theoretical curves of residual velocity versus initial impact velocity fitted by the Recht-Ipson (R-I) function (Recht and Ipson, 1963) and calculated by the current analytical model.

The observation of Fig. 6 indicates that the measured ballistic limit point (fitted by the R-I function) is 109.4 m/s, and the theoretical one is 107.5 m/s, with the related calculation error about -1.74% , which is within an acceptable range for engineering applications. Moreover, the calculated curve of residual velocity is slightly lower than the measured one, indicating that the proposed model provides a slightly conservative estimation of residual velocity. In addition, both calculated and

Table 4

The related instruments adopted for the high-velocity impact experimental system.

Instrument	Model	Manufacturer and specification
High-pressure gas cylinder	QF-2	Linyi Hedong District Sodi Hardware Tool Wholesale Firm, China, Maximum allowable working pressure (MAWP): 14 MPa, volume: 40L
Pressure reducing valve	KH41-316L	Nanjing Kehang Experimental Instrument Co., Ltd., China, MAWP: 21MPa
Inlet valve	ZYS-2	Anhui Jiangnan pump and valve group Co., Ltd., China, MAWP: 15 MPa
Launch gas storage device	–	Customized model, Volume: 9.42L
Projectile launch tube	–	Customized model, Inner wall polished, inner diameter: 20 mm, outer diameter: 30 mm, length: 120 mm
Projectile	–	Customized model, Diameter of hemispherical head: 19.6 mm, length of cylindrical nylon body: 34 mm, total mass: 22g
Infrared receiving phototransistor array	PT204	EVERLIGHT, China, Wavelength of peek sensitivity: 940 nm
Infrared emitting phototransistor array	IR204	EVERLIGHT, China, Wavelength of peek sensitivity: 940 nm
Illuminator	EFII-200BI	JINBEI, China, Aperture value: F32.2, maximum power of the illuminator: 200 Ws
High-speed camera	ISP504-ISP	Hefei Zhongke Junda Vision Technology Co., Ltd., China, Maximum frames: 13500 FPS

measured curves of residual velocity show a slightly convex shape at relatively low impact velocities, which become more linear at higher velocities, suggesting that the energy dissipation mechanisms change with the increment of impact velocity. Meanwhile, the energy absorption effect of the HAHC-CCS specimens becomes less effective at a high impact velocity of the projectile owing to the gradually decreasing slope of the calculated and measured curves of residual velocity.

The experimental and calculated residual velocities of the projectile after Specimens I to V are perforated are listed in Table 5 for ease of comparison. It can be seen that the current model provides a reasonably good prediction of the residual velocity of the projectile for each tested specimen, with a maximum calculation error being -5.6% .

In addition, by taking Specimens III, IV, and V as examples, the related damage photographs of the inner and outer skins after they are perforated are provided in Fig. 7. One can clearly see that a progressively mitigated damage phenomenon happening on the outer skins of these specimens, which is primarily due to fiber tension failure and delamination damage. Also, the inner skins exhibit relatively neat

fractures, indicating that their main damage mode is compression failure. However, the fracture fibers of the outer skin display a relatively slender shape caused by tensile failure. Moreover, the calculated damage mode of the inner skin is mainly compression failure, whereas the failure mode of the outer skin is mainly tensile failure, which has a good correlation with the measured damage modes. It should be noted that it is difficult to examine the internal damage of the penetrated HAHC due to space limitation, but the calculated and measured impact damage areas of the inner and outer skins of these specimens are compared, as listed in Table 6. Here, these damage areas are indirectly calculated

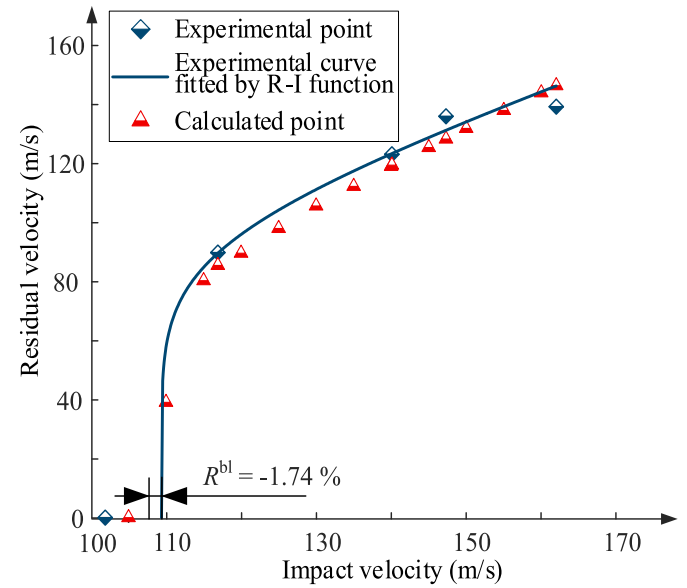


Fig. 6. Comparison of the experimental and predicted residual velocity curves and ballistic limit of the specimens.

Table 5

Comparison of the experimental and calculated residual velocities of the projectile after Specimens I to V are perforated.

Initial impact velocity (m/s)	Residual velocity (m/s)		Error (%)
	Experimental	Calculated	
101.9	0	0	0
116.9	90.1	85.7	-4.9
140.1	123.5	119.8	-3.0
147.3	136.4	128.7	-5.6
162.0	139.7	146.8	5.1

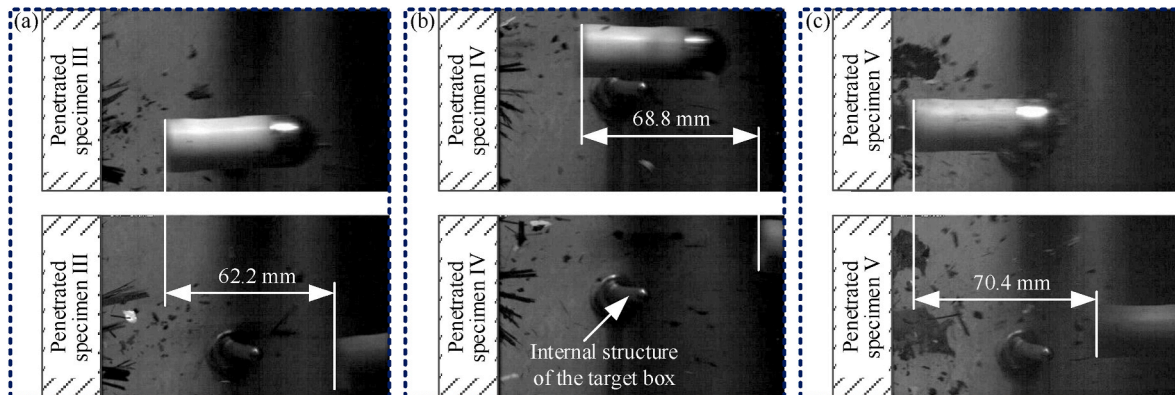


Fig. 5. The fighting distances of the projectile between the continuous photographs after Specimens III, IV, and V are perforated with the initial impact velocities of (a) 140.1, (b) 147.3, and (c) 162.0 m/s.

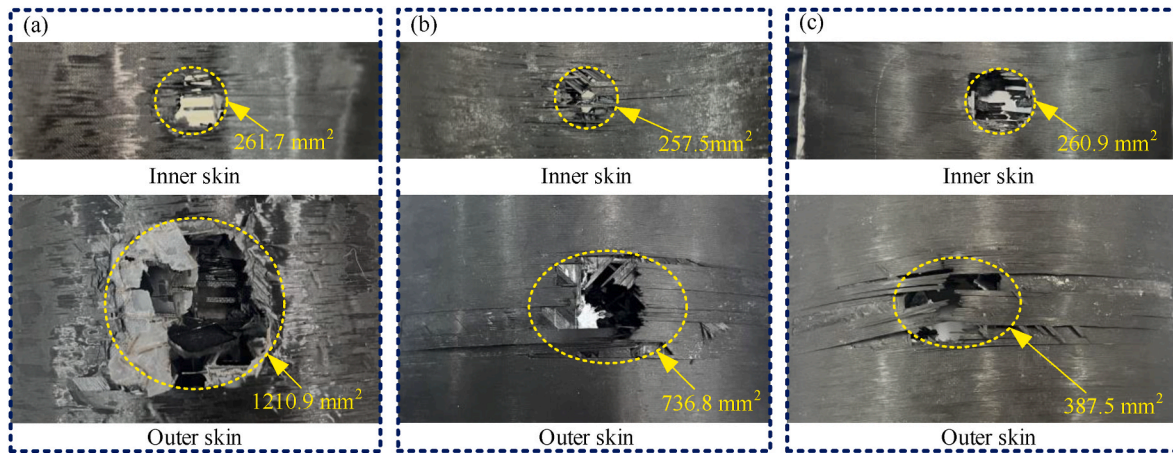


Fig. 7. The impact damage photographs of the inner and outer skins of (a) Specimen III, (b) Specimen IV, and (c) Specimen V.

Table 6

Comparison of the experimental and calculated damage areas of Specimens III, IV, and V.

Initial impact velocity (m/s)	Inner skin damage area (mm ²)		Error (%)	Outer skin damage area (mm ²)		Error (%)
	Experimental	Calculated		Experimental	Calculated	
140.1	261.7	270.9	3.5	1210.9	1162.2	-4.0
147.3	257.5	264.2	2.6	736.8	778.3	5.6
162.0	260.9	251.8	-3.4	387.5	360.1	-6.1

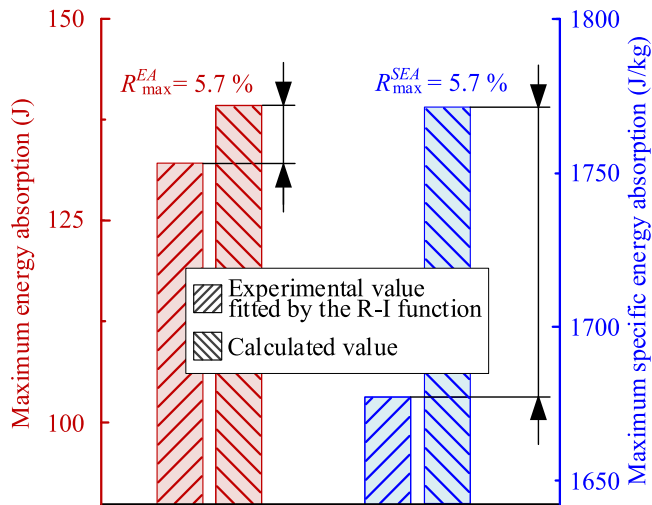


Fig. 8. Comparison of the (a) maximum energy absorption and (b) maximum specific energy absorption obtained from experiments and the present model.

using the image processing software, ImageJ, a Java-based program (Schroeder et al., 2021).

It can be seen from Fig. 7 and Table 6 that the largest calculation errors of impact damage areas of the inner and outer skins are 3.5 % and -6.1 %, respectively, and both fall within a tolerable range. Also, the extent of damage on the outer skins of these specimens is more severe than that on the inner skins, which indicates that the outer skins of the specimens experience greater damage in the same impact conditions. On the one hand, the photograph data reveal a clear decrease in the extent of damage to the outer skins as the initial impact velocity increases from 140.1 to 162.0 m/s, with the shape of the damaged area of these specimens becoming more elongated and irregular. This is because the outer skins of the specimens experience more delamination and fiber breakage

damage, resulting in a more elongated and irregular shape in the damaged area, in comparison to the inner skins, which is consistent with the previous studies on the high-velocity impact of sandwich structures with composite skins (Aryal et al., 2019). On the other hand, the damaged area on the inner skin shows little variation, with the shape of the damaged region remaining circular. In addition, the reduction in the outer skin damaged area with an increment of initial impact velocity is likely due to the strain-rate dependence of composite skins (Ma et al., 2022), which can lead to an increase in their toughness and strength, since the strain-rate effect becomes more evident as initial impact velocity of the projectile rises. Another reason may be that a severe impact phenomenon with a higher impact velocity can generate more micro-cracks in the core of the specimens tested, which will impede the propagation of structural cracks and thus absorb impact energy by the specimens. A similar view can also be found in Refs. (MADKE and CHOWDHURY, 2020; CANDAS et al., 2021), in which the measured results indicate that a core with the negative Poisson ratio property plays an important role in enhancing energy absorption capacity.

Next, Fig. 8 illustrates the calculation errors of the maximum values of impact energy absorption and specific energy absorption under different initial velocities obtained from experiments and the present model. It can be observed that the predicted results agree well with the measured ones, with both of the calculation errors being less than 5.7 %. Hence, the proposed model is reliable enough to provide a reasonable prediction for high-velocity impact parameters of composite cylindrical shells with an auxetic honeycomb core.

Finally, it is necessary to investigate the sources of the calculation errors of the current model, which probably come from the following factors: 1) the lack of consideration of the relative size and position of the projectile and the HAHC of the studied structure; 2) the ignorance of the constitutive relationship of the adhesive film of epoxy resins between the skins and the core; 3) the neglect of the influence of energy absorption mechanisms caused by frictional force and heat generated between projectile and the structure; 4) the simplification of the failure criterion in which the influence of structural characteristics of the HAHC on the failure mode is ignored.

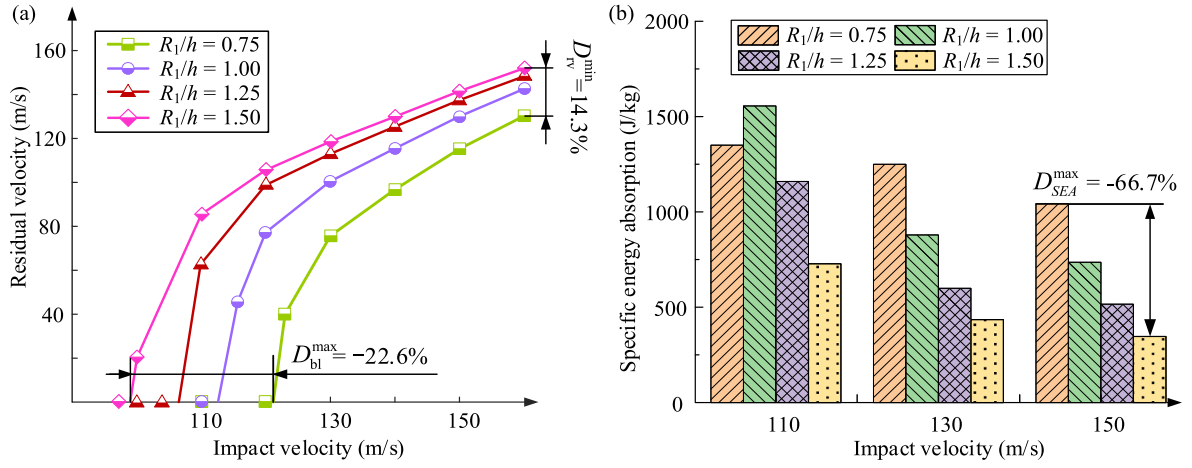


Fig. 9. Influence of the ratio of the radius of the projectile to the overall thickness on (a) the structural residual velocity and (b) specific energy absorption.

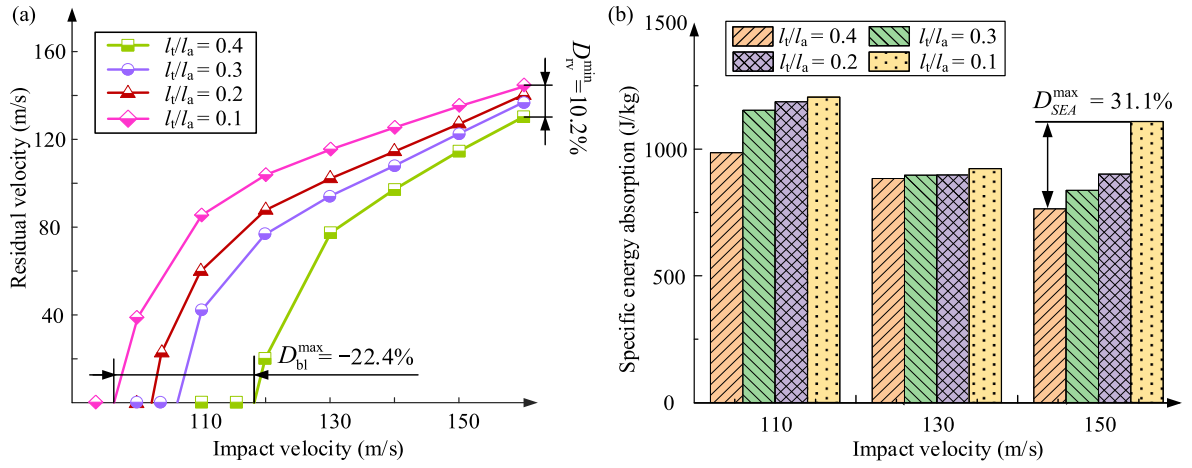


Fig. 10. Effect of the ratio of the rib thickness to the length of the inclined cell rib of the HAHC on (a) the structural residual velocity and (b) specific energy absorption.

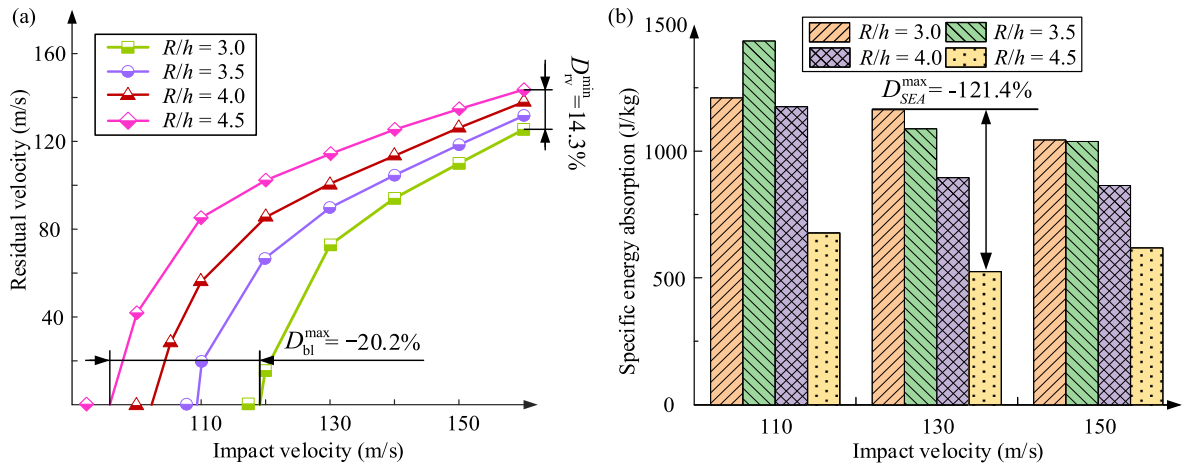


Fig. 11. Influence of the ratio of the structural radius to the overall thickness on (a) the structural residual velocity and (b) specific energy absorption.

5. Parametrical study

5.1. Influence of the ratio of the radius of the projectile to the overall thickness

The influence of the ratio of the radius of the projectile to the overall thickness, R_1/h , on the high-velocity impact parameters of the HAH-CCS structure is firstly investigated, as shown in Fig. 9, in which the minimum deviation values of ballistic limit, residual velocity, and specific energy absorption are represented as D_{bl}^{max} , D_{rv}^{min} , and D_{SEA}^{max} respectively. Note that the calculations make use of the geometric dimensions, material parameters, fitting coefficients, initial parameters of the iterative algorithm, and other important parameters declared in Section 4.3, except that the mass of the projectile varies from 20 to 26 g with the change value of R_1 .

One can find out that as R_1/h rises from 0.75 to 1.5, the residual velocity curve exhibits an upward trend with $D_{rv}^{min} = 14.3\%$, whereas the ballistic limit decreases with $D_{bl}^{max} = -22.6\%$. Meanwhile, the energy absorption curve exhibits a declining trend with $D_{SEA}^{max} = -66.7\%$, suggesting that increasing energy absorption efficiency is best achieved with a modest R_1/h . These intricate variations in high-velocity impact characteristics may be owing to the increased perforation capacity of the projectile since the mass M rises while h remains constant. Another reason is that the contact area R_2 rises with the increase of the radius of the projectile, resulting in a larger delamination area of the structure and more consumption of impact energy. For such a shell, it is therefore advised to employ a small ratio of the projectile radius to the total thickness in order to meet the high-impact energy absorption demand.

5.2. Influence of the ratio of the rib thickness to the length of the inclined cell rib of the core

Fig. 10 presents the effect of the ratio of the rib thickness to the length of the inclined cell rib, l_r/l_a , on the high-velocity impact parameters of the HAH-CCS structure. In the above investigations, the same geometric dimensions, material parameters, fitting coefficients, initial parameters of the iterative algorithm, and other key parameters adopted in Section 4.3 are utilized, except that $R_1/h = 1.5$ is chosen, and the structural mass is set from 42 to 135 g with a changing value of l_a .

One can clearly see from Fig. 10 that with the decline of l_r/l_a , the value of ballistic limit reduces with $D_{bl}^{max} = -22.4\%$, but the residual velocity and the energy absorption results of the HAH-CCS indicate an uptrend with $D_{rv}^{min} = 10.2\%$ and $D_{SEA}^{max} = 31.1\%$. Note that similar findings are also reported in Ref. (Tiwari and Khaire, 2022). The changes in the above high-velocity impact parameters may be caused by the decreased equivalent Young's and shear moduli $E_1^C, E_2^C, G_{12}^C, G_{13}^C, G_{23}^C$ and the increased value of negative Poisson's ratio ν_{12}^C (from -37.8 to -0.11) and ν_{21}^C (from -0.52 to -0.06) of the HAH in Eq. (8). As a result, there is a downtrend change in the overall stiffness and a slight reduction in equivalent Poisson's ratio of the structure, making it easier for the projectile to perforate the structure. In addition, the drop in l_r/l_a also results in a larger cell size or lower density of HAH per unit volume, leading to a decreased mass of the structure, which explains the reason for an uptrend in specific energy absorption. Therefore, to maintain good high-velocity impact resistance, a large ratio of the rib thickness to the length of the inclined cell rib is recommended.

5.3. Influence of the ratio of the structural radius to the overall thickness

The influence of the ratio of the structural radius to the overall thickness, R/h , on the impact properties of the HAH-CCS is investigated when initial impact velocity changes, as displayed in Fig. 11. Here, the identical geometric dimensions, material parameters, fitting coefficients, initial parameters of the iterative algorithm, and other key parameters used in Section 4.3 are adopted, except that $R_1/h = 1.5$ and

$l_r/l_a = 0.2$ are chosen, and the thickness of each layer varies proportionally with the total thickness.

As demonstrated by the observation of Fig. 11, as the value of R/h rises, the value of residual velocity indicates an increasing trend with $D_{rv}^{min} = 14.3\%$, whereas the structural energy absorption and ballistic limit results decline with $D_{SEA}^{max} = -121.4\%$ and $D_{bl}^{max} = -20.2\%$, which proves that a small R/h of such a shell is advantageous for improving specific energy absorption efficiency. The Poisson's ratio and equivalent moduli of the structure in Eq. (18) fall as R/h increases, which may be the result of these complex changes in dynamic parameters as well as the decreasing thickness of each layer and the mass m_s of the structure. Ref. (MADKE and CHOWDHURY, 2020) comes to a similar conclusion, showing that increasing the thickness of the shell can increase its impact resistance. Therefore, a small ratio of the structural radius to thickness is advised to help obtain better anti-impact performance of such a shell.

6. Conclusions

In this study, a novel analytical model of the HAH-CCSs is established to predict the high-velocity impact parameters under the internal impact of a hemispherical head projectile. The proposed model is validated through the comparison with the experimental data when the impact velocity increases from 101.9 m/s to 162.0 m/s. Since the influences of impact failure modes of the core and skins are taken into account, a good prediction of the impact features is captured, with the related prediction error being less than 6.1 % in residual velocity, ballistic limit, damage areas, impact energy absorption, etc.

Based on the measured and predicted impact results of the studied shells, one can find out that as the initial impact velocity increases, the damage areas of both the inner and outer skins of the shells decrease, with the outer skins exhibiting larger damage areas and more elongated shapes than the inner skins. Additionally, to enhance the high-velocity impact resistance of the HAH-CCS structure, it is recommended to adopt a small ratio of the projectile radius to total thickness, a large ratio of the rib thickness to the length of the inclined cell rib, and a small ratio of the structural radius to thickness for such a shell.

The model developed in this study can assist engineers and researchers in designing lightweight composite cylindrical shells with an advanced auxetic core to possess good impact resistance. Also, the related assumptions and treatment techniques for dynamic constitutive models of such composite structures can be applied to the finite element simulation process.

CRedit authorship contribution statement

Hui Li: Writing – review & editing, Validation, Investigation. **Yichen Deng:** Writing – original draft, Visualization, Formal analysis. **Zelin Li:** Software. **Jin Zhou:** Validation, Resources, Investigation. **Haijun Wang:** Conceptualization. **Shaoming Wang:** Visualization. **Haiyang Zhang:** Methodology. **Hang Cao:** Project administration. **Xiangping Wang:** Supervision, Data curation, Conceptualization. **Qingkai Han:** Supervision, Conceptualization. **Zhongwei Guan:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability

Data will be made available on request.

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Appendix A

In Eq. (10), $T^{(l)}$ can be expressed as:

$$T^{(l)} = \begin{bmatrix} \cos^2 \theta_F^{(l)} & \sin^2 \theta_F^{(l)} & 0 & 0 & 0 & \sin \theta_F^{(l)} \cos \theta_F^{(l)} \\ \sin^2 \theta_F^{(l)} & \cos^2 \theta_F^{(l)} & 0 & 0 & 0 & -\sin \theta_F^{(l)} \cos \theta_F^{(l)} \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos \theta_F^{(l)} & -\sin \theta_F^{(l)} & 0 \\ 0 & 0 & 0 & \sin \theta_F^{(l)} & \cos \theta_F^{(l)} & 0 \\ -\sin 2\theta_F^{(l)} & \sin 2\theta_F^{(l)} & 0 & 0 & 0 & \cos^2 \theta_F^{(l)} - \sin^2 \theta_F^{(l)} \end{bmatrix} \quad (A1)$$

Appendix B

In Eq. (13), B_{mn} , D_{mn} , E_{mn} , F_{mn} , and H_{mn} have the following expression:

$$(B_{mn}, D_{mn}, E_{mn}, F_{mn}, H_{mn}) = \bar{Q}_{mn} \int_{z_1}^{z_{N+1}} (z, z^2, z^3, z^4, z^6) dz \quad (B1)$$

Appendix C

In Eq. (26), G_{II} can be expressed as:

$$G_{II} = \frac{9Fw_{\max}L_{\text{del}}^2}{2L \left(\frac{L^2}{4} + 3L_{\text{del}}^3 \right)} \quad (C1)$$

where L_{del} is the crack length, which can be taken as three times the diameter of the projectile.

Appendix D

In Eq. (28), T_u can be stated as:

$$T_u = U - U^b \quad (D1)$$

where U^b represent the total strain energy before the bullet contacts the innermost layer, which can be induced by various deformation modes, including U_A^b , U_B^b , U_D^b , U_E^b , U_F^b , and U_H^b . Their relations can be expressed in terms of the displacement and rotation w_0^b , ϕ_1^b , and ϕ_2^b before the bullet contacts the innermost layer, which can be derived as:

$$U_A^b = \iint_A \left\{ \begin{aligned} & A_{11} \frac{1}{4} \left(\frac{\partial w_0^b}{\partial x} \right)^4 + 2A_{12} \left[\frac{1}{2} \frac{w_0^b}{R} \left(\frac{\partial w_0^b}{\partial x} \right)^2 + \frac{1}{4} \left(\frac{\partial w_0^b}{\partial x} \frac{\partial w_0^b}{\partial \theta} \right)^2 \right] + 2A_{16} \left[\frac{1}{2} \left(\frac{\partial w_0^b}{\partial x} \right)^2 \frac{\partial w_0^b}{\partial x} \frac{\partial w_0^b}{\partial \theta} \right] \\ & + A_{22} \left[\left(\frac{w_0^b}{R} \right)^2 + \frac{1}{4} \left(\frac{\partial w_0^b}{\partial \theta} \right)^4 + \frac{w_0^b}{R} \left(\frac{\partial w_0^b}{\partial \theta} \right)^2 \right] + 2A_{26} \left[\frac{w_0^b}{R} \frac{\partial w_0^b}{\partial x} \frac{\partial w_0^b}{\partial \theta} + \frac{1}{2} \left(\frac{\partial w_0^b}{\partial \theta} \right)^2 \frac{\partial w_0^b}{\partial x} \frac{\partial w_0^b}{\partial \theta} \right] \\ & + A_{66} \left(\frac{\partial w_0^b}{\partial x} \frac{\partial w_0^b}{\partial \theta} \right)^2 + A_{44} \left[\left(\frac{\partial w_0^b}{\partial \theta} \right)^2 + \phi_2^{b2} + 2 \frac{\partial w_0^b}{\partial \theta} \phi_2^b \right] \\ & + 2A_{45} \left[\frac{\partial w_0^b}{\partial x} \frac{\partial w_0^b}{\partial \theta} + \phi_2^b \frac{\partial w_0^b}{\partial x} + \phi_1^b \frac{\partial w_0^b}{\partial \theta} + \phi_1^b \phi_2^b \right] + A_{55} \left[\left(\frac{\partial w_0^b}{\partial x} \right)^2 + \phi_1^{b2} + 2 \frac{\partial w_0^b}{\partial x} \phi_1^b \right] \end{aligned} \right\} dA \quad (D2a)$$

(D2b)

(D2c)

(D2d)

(D2e)

(D2f)

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